MIL-T-46659A(MU)
31 January 1968

SUPERSEDING
MIL-T-46659 (Ord)
4 May 1962

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M5A2, ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M5A2, Assembly, loaded with Cartridge, Impulse, M73, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MS28741 - Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS
Munitions Command
F8594746 - Thruster, Cartridge Actuated, M5A2, Assembly
C8797741 - Container, Ammunition, Fiber M347 for Thruster Cartridge Actuated, M20A1, XM25 or M5A2 (See 5.1)
C8797742 - Box, Packing, Ammunition, for Thruster, Cartridge Actuated, M20A1, XM25 or M5A2, in Fiber Container M347 (See 5.1)

FSC 1377

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2.1 DRAWINGS (Cont'd)

IEL-8594746 - Inspection Equipment List for Thruster Cartridge Actuated, M5A2, Assembly

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specification.

3.2 Torque.- Torque requirements shall be as specified in Table I. No movement shall occur upon application of the breakaway torque.

TABLE I

Components	A Seating or Bottoming (Inch-Pounds)	B Breakaway, min. (Inch-Pounds)
Head and Cartridge Retainer after Insertion of Cartridge	300 ± 20	200
Head and Body	600 ± 20	400

3.3 Assembly.

3.3.1 Trunnion rotation.- Upon completion of assembly the trunnion shall be free to rotate 360 degrees.

3.4 Residual magnetism.- The thruster shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.5 Functional requirements.

3.5.1 The thruster shall not fail to fire.

3.5.2 The thruster shall function without mechanical failure (see 6.1).

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3.5.3 No load.- The thruster fired with no load on the piston shall complete a stroke of five inches, minimum. The piston shall not separate from the body of thruster (See 6.2).

3.5.4 Load.- The thruster fired under load shall perform as follows:

3.5.4.1 Move a 500-pound carriage on wheels, horizontally, through a distance of five inches, minimum, starting from zero-inches stroke; overcome an initial restraining force of 4000 pounds located at zero-inches stroke and complete a piston stroke of five inches, minimum (See 6.3).

3.5.5 Temperature.- The thruster shall comply with all the ballistic requirements throughout the temperature range of minus 65° to plus 165°F.

3.5.6 Initiation.- The thruster shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4, as specified in MS28741, at $70^{\circ} \pm 5^{\circ}$F.

3.6 Workmanship.- Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified therein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

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4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots, lot size, and presentation of lots shall be in accordance with the "Submission of Products", provisions of MIL-STD-105.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 - 10000	200	2	3	3	4

4.2.2.1 Cartridge, Impulse, M73 Drawing D8594157 covering a detail of Drawing F8594746.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Head sub-assembly, consisting of the following: Head 8594749 Cartridge Retainer 8594483, Guide 8593863, Firing Pin 8593864, Shear Pin 8593284 and O-ring MS29513-12 covering a detail of Drawing F8594746.

CRITICAL	
1. Cartridge loose or missing	Manual and Visual
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Thread damaged or not full	Visual
103. Protective finish inadequate or defective	Visual
104. Metal defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly Drawing F8594746 less Shipping Plug 8593299.

CRITICAL	
1. Shear pin not engaged in firing pin	Gage
2. Foreign matter in gas port	Visual
MAJOR	
101. Thread damaged	Visual
MINOR	
201. Protective finish inadequate or defective	Visual

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4.2.2.4 Assembly, Drawing F8594746.

Defect	Method of Inspection
CRITICAL	
1. Improper assembly	X-ray
2. Any part missing as detected by X-ray	X-ray
3. Metal defective	Visual
MAJOR	
101. Trunnion not free to rotate 360 degrees	Manual
102. Foreign matter	Visual
103. Marking missing, incorrect or illegible	Visual
MINOR	
201. Total length	Measure
202. Location of labels incorrect	Visual
203. Shipping cap or plug missing	Visual
204. Protective finish inadequate or defective	Visual
205. Varnish missing or inadequate	Visual

4.2.2.5 Fiber Container, Drawing C8797741.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

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4.2.2.6 Box, Packing, Drawing C8797742.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.

4.2.3.1.1 Head and cartridge retainer.- Each head and cartridge retainer subassembly shall be subjected to a breakaway torque application. Failure of any subassembly to comply with the requirement of Column B, Table I, in 3.2 shall be cause for rejection of the subassembly.

4.2.3.1.2 Head and body.- Each head and body assembly shall be subjected to a breakaway torque application. Failure of any assembly to comply with the requirement of Column B, Table I in 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.- Each thruster shall be subjected to the test for excessive residual magnetism. Failure of any thruster to comply with the requirement of 3.4 shall be cause for rejection of the thruster.

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4.2.3.3 Functional tests.- The following tests will be performed by the Government.

4.2.3.3.1 Ballistics.- Thirty units shall be tested to determine compliance with the requirements of 3.5.1 through 3.5.5. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

TABLE III

Sample Size	Method	Temperature
9	Load (See 3.5.4)	-65° to -70°F
9	Load (See 3.5.4)	+65° to +75°F
9	Load (See 3.5.4)	+160° to +165°F
1	No Load (See 3.5.3)	-65° to -70°F
1	No Load (See 3.5.3)	+65° to +75°F
1	No Load (See 3.5.3)	+160° to +165°F

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment in accordance with IEL-8594746.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, capable of measuring the torque requirements as specified in Table I shall be used with an appropriate adapter to determine the adequacy of the joints.

4.3.2 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be followed for each 90 degree rotation of the thruster.

4.3.3 Functional tests.

4.3.3.1 Ballistics.- The test samples shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperatures.

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4.3.3.1.1 Temperature conditioning.- A simulated thruster and cartridge with a temperature sensitive element, placed in the dummy cartridge, shall be stored in a temperature conditioning chamber at the same time as the samples. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required test temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with Drawings C8797741 and C8797742, except that the markings shall reference Thruster, M5A2, with M73 Cartridge.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 No load.- Permanent yielding of the trunnion mounting lugs and piston shall not be cause for rejection when thrusters are tested under "no load" condition.

6.3 Load.- The expected performance of the thruster under conditions simulating those existing in aircraft shall be referred to as "load" condition in this specification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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